Lesson notes · for you

Databases and SQL: unit check

Data engineering · Level 3 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Tables, keys and types
- Questions in SQL
- Joining tables
- Fast answers: indexes

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (15 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (8 min): Reveal each answer and ask one team to explain why it's right.

Discussion

- Which app's database would you like to see inside?
- When would a left join save you from a wrong answer?
- Which column would you index first, and why?

Check yourself

- One query I could now write that I couldn't before, and what it answers.

Try it this week
Sketch the tables of an app you use, write one JOIN you'd run on them, and say which column you'd index.

Worksheet

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Name _____ Class _____ Date _____

1 **Which column makes the best primary key for an app's users?**

◯ A user id given once and never reused ◯ The user's name ◯ The user's country

2 **A foreign key points to a row in another table.**

◯ True ◯ False

3 **Put the parts of a query in the order SQL needs them.**

☐ LIMIT
☐ ORDER BY
☐ GROUP BY
☐ WHERE
☐ FROM
☐ SELECT

4 **A table has 12 rows; one mark is NULL. What does this return?**

```
SELECT COUNT(*), COUNT(mark) FROM marks;
```

◯ 12 and 11 ◯ 12 and 12 ◯ 11 and 11

5 **Match each join to what it does with a member who never borrowed.**

JOIN (inner)	• • • •	Shows only members like them
LEFT JOIN members to loans	• • • •	Keeps them, with NULL for the book
LEFT JOIN … WHERE book IS NULL	• • • •	Leaves them out